

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1521

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Total Marks: 50

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Code of Conduct

I Harindra (19252025) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

CO2. Assessment - 1

Q. Compare Type 1 and Type 2 Hypervisors better for mission-critical applications?

Type-1 Hypervisor:-

- Runs directly on hardware.
- Better performance.
- More performance.
- Easy to manage in data

Type 2 Hypervisor:-

- Runs on top of an operating system.
- Slightly slower.
- Less secure because it depends on the host OS.
- mainly used for testing and learning.
- Type 1 hypervisor provide better speed and performance.
- they have stronger security because there is no host operating system.
- They are easier to manage in large cloud environments.

27

How does virtualization isolate workloads?

Name two attack surfaces.

Ans:-

virtualization creates separate virtual machines (VM's) on the same physical server.

Benefits:-

- Each VM works independently.
- problems in one VM do not affect others.
- Finance, healthcare, and student portal data remain isolated and secure.

Two major attack surfaces:-

- Hypervisor attack → An attacker tries to exploit the hypervisor.
- VM escape attack → A malicious VM tries to access another VM or the host system.

⇒ Analyze the VM host snapshot and suggest corrective actions.

→ CPU = 92%.

→ Memory = 88%.

→ Storage I/O latency = 35ms.

→ VM boot time = 170 sec..

likely bottlenecks:-

→ CPU usage is very high, causing slow processing.

→ Memory is almost full, reducing performance.

→ High storage latency slows down data access.

→ Long boot time means the server is overloaded.

corrective actions:-

→ Add more CPU resources.

→ Increase RAM.

→ Upgrade to faster SSD storage.

→ Balance VM's across multiple servers.

4/5

How does a software defined (SDDC) improve agility?

Data center

- compute virtualization.
- uses hardware efficiently.

storage virtualization:-

- storage can be allocated as needed.
- Easy to expand storage.

Network virtualization:-

- creates virtual networks.
- improves security and flexibility.

5/5

Multi-cloud federation fails due to identity mismatch. Give a secure solution.

problem:-

Different cloud providers use different user identities, making authentication difficult.

Solu:-

- use identity federation.
- Enable single - on (SSO).
- use secure authentication methods like OAuth or SAML.
- Apply multi - factor authentication (MFA).
- Encrypt user data to protect privacy.